

GGL90 Validation Report

MITgcm vs Python Port Comparison (single-step replay)

Generated: 2026-09-27 05:56:23

MITgcm Reference:

File: mitgcm_ggl90_outputs_1D_ocean_ice_column_11000.nc
UUID: 1c8c7e52-3041-4f49-ae2b-841705d2d7a3

Python Port Output:

File: python_ggl90_outputs_1D_ocean_ice_column_11000.nc
UUID: 1c8c7e52-3041-4f49-ae2b-841705d2d7a3

Provenance: ☐ UUIDs match

Timesteps: 11000

Grid: 1 × 1 × 23

Fields compared: visc_az, diff_kz, mixing_length, tke_after
(GGL90 single-step replay: Python fed MITgcm's own captured
tke_before(t); only the one-step update is compared, isolating
agreement on the physics formula from trajectory drift.)

GGL90 Field Statistics

Wet-cell comparison (NaN-excluded below-seafloor/above-ice-shelf cells)

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visc_az  (Vertical eddy viscosity (K_m), m^2/s)
          Status:  EXCELLENT
Median |diff|: 0.0000e+00  P95 |diff|: 6.9389e-18  Max |diff|: 1.7627e-09
Median rel:   0.0000e+00  P95 rel:   4.0110e-15  Max rel:   7.9362e-07
          Fraction >1% rel error: 0.0000%
          Points compared: 253000

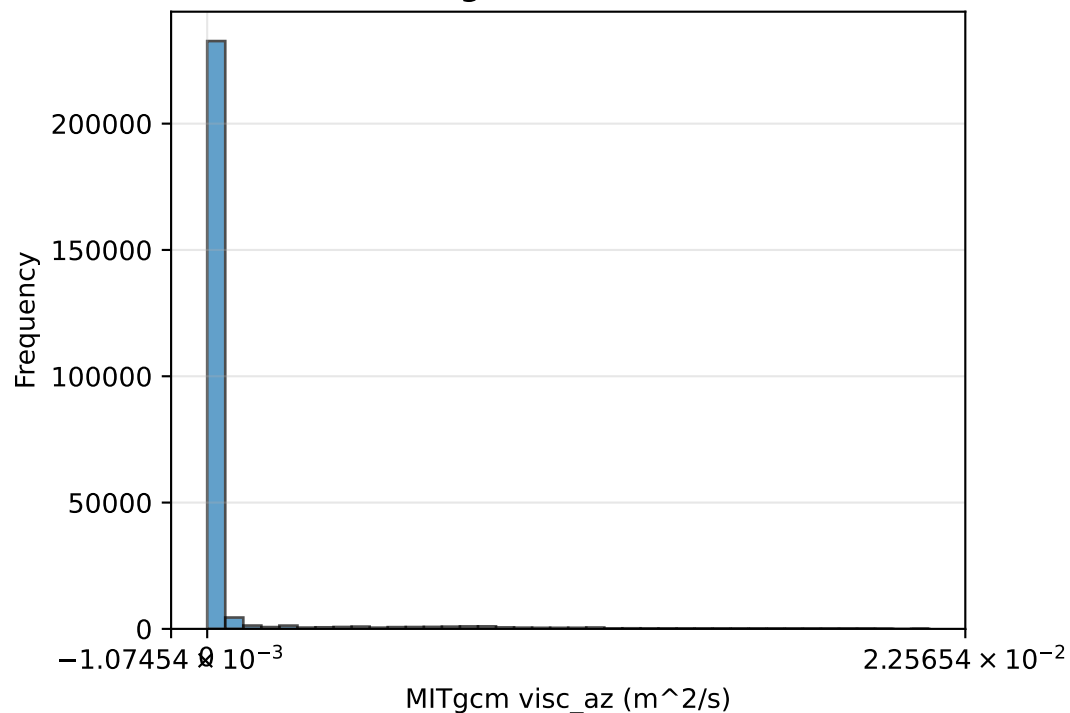
diff_kz  (Vertical eddy diffusivity (K_h), m^2/s)
          Status:  EXCELLENT
Median |diff|: 0.0000e+00  P95 |diff|: 3.1995e-17  Max |diff|: 1.7627e-10
Median rel:   0.0000e+00  P95 rel:   1.6326e-11  Max rel:   7.9362e-07
          Fraction >1% rel error: 0.0000%
          Points compared: 253000

mixing_length (GGL90 mixing length (L), m)
          Status:  EXCELLENT
Median |diff|: 2.5778e-15  P95 |diff|: 4.5353e-12  Max |diff|: 5.5741e-05
Median rel:   3.8561e-15  P95 rel:   1.6326e-11  Max rel:   7.9362e-07
          Fraction >1% rel error: 0.0000%
          Points compared: 253000

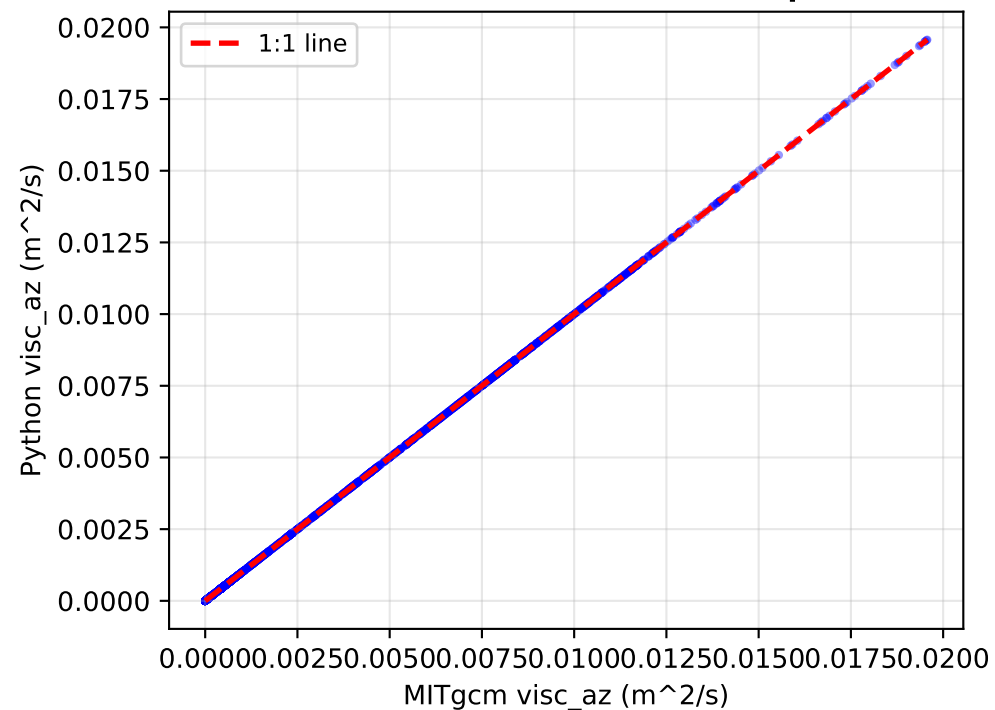
tke_after (TKE at t+1 (single-step replay), m^2/s^2)
          Status:  EXCELLENT
Median |diff|: 0.0000e+00  P95 |diff|: 1.6139e-17  Max |diff|: 6.6155e-15
Median rel:   0.0000e+00  P95 rel:   1.8797e-12  Max rel:   5.1386e-09
          Fraction >1% rel error: 0.0000%
          Points compared: 253000
```

visc_az (Vertical eddy viscosity (K_m)) -- MITgcm vs Python

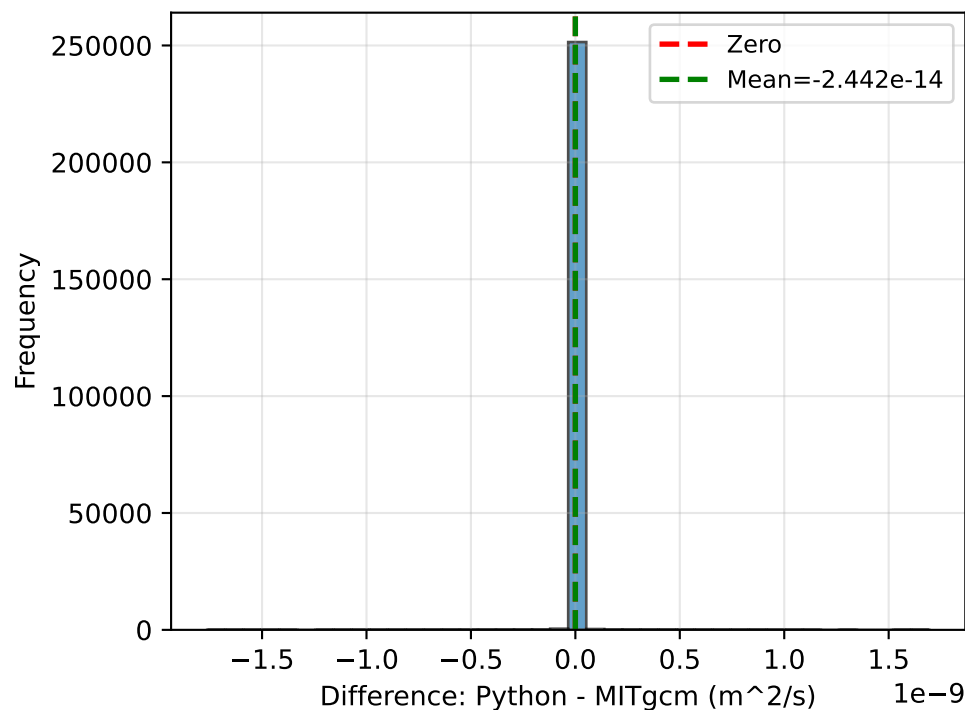
MITgcm Distribution



Scatter Plot (20000 of 253000 pts)



Difference Distribution



visc_az Statistics:

Median |diff|: 0.0000e+00
P95 |diff|: 6.9389e-18
Max |diff|: 1.7627e-09

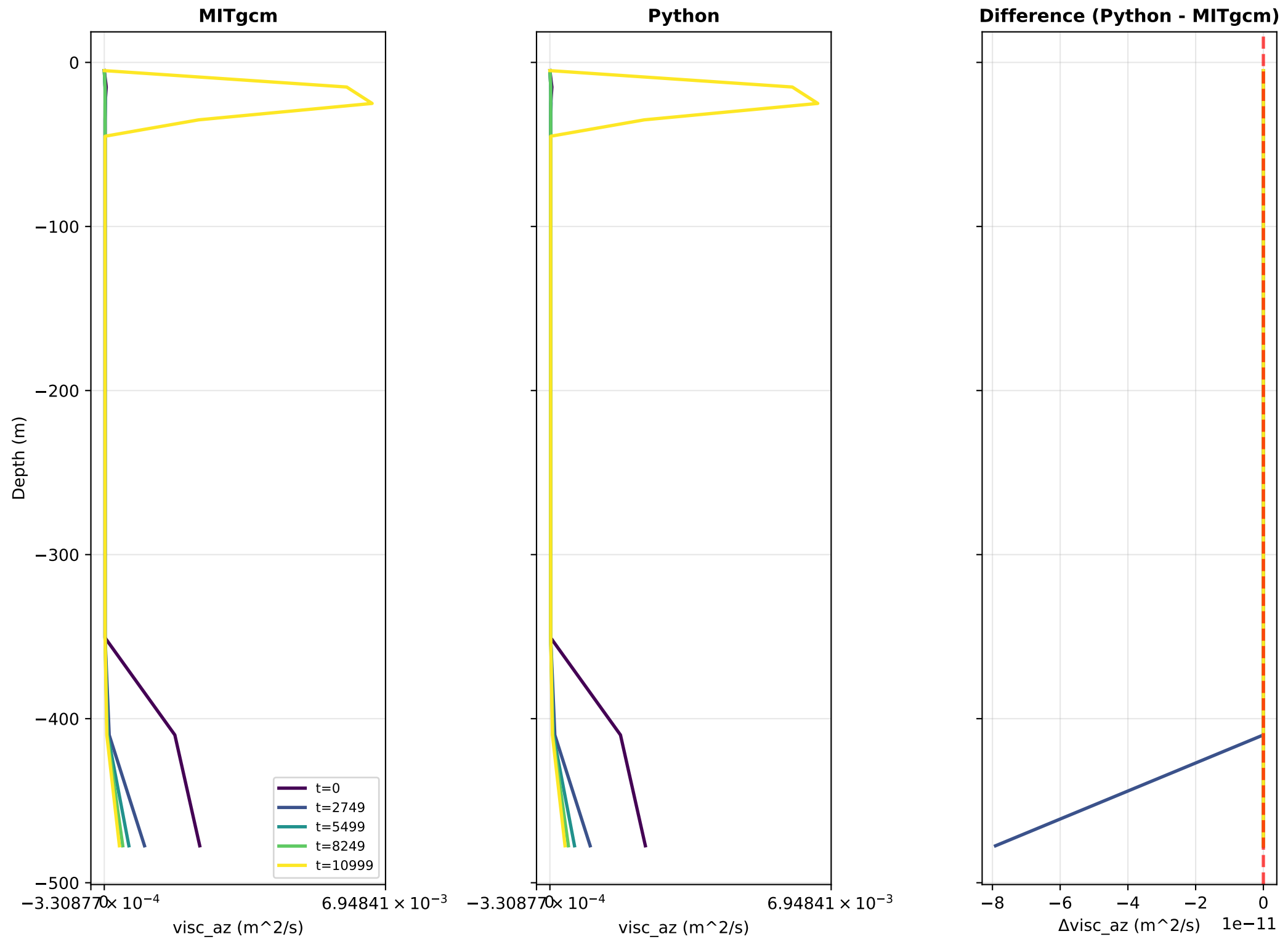
Median rel: 0.0000e+00
P95 rel: 4.0110e-15
Max rel: 7.9362e-07

Fraction >1% rel: 0.0000%
Points compared: 253000

Assessment:

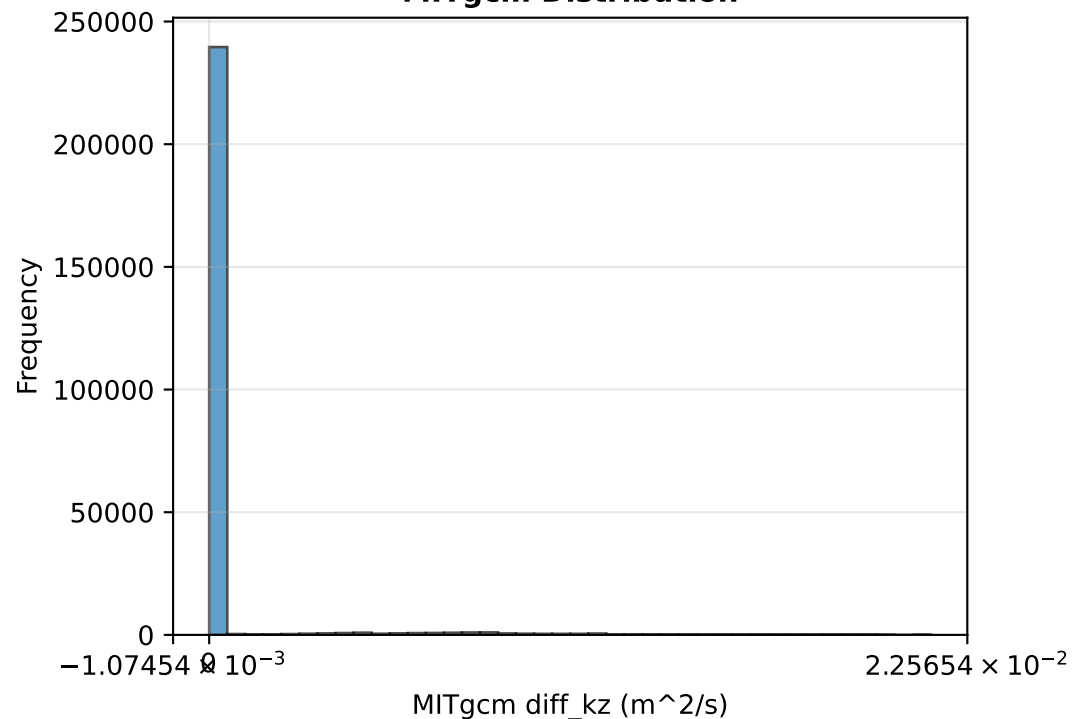
EXCELLENT

visc_az Vertical Profiles (Vertical eddy viscosity (K_m))

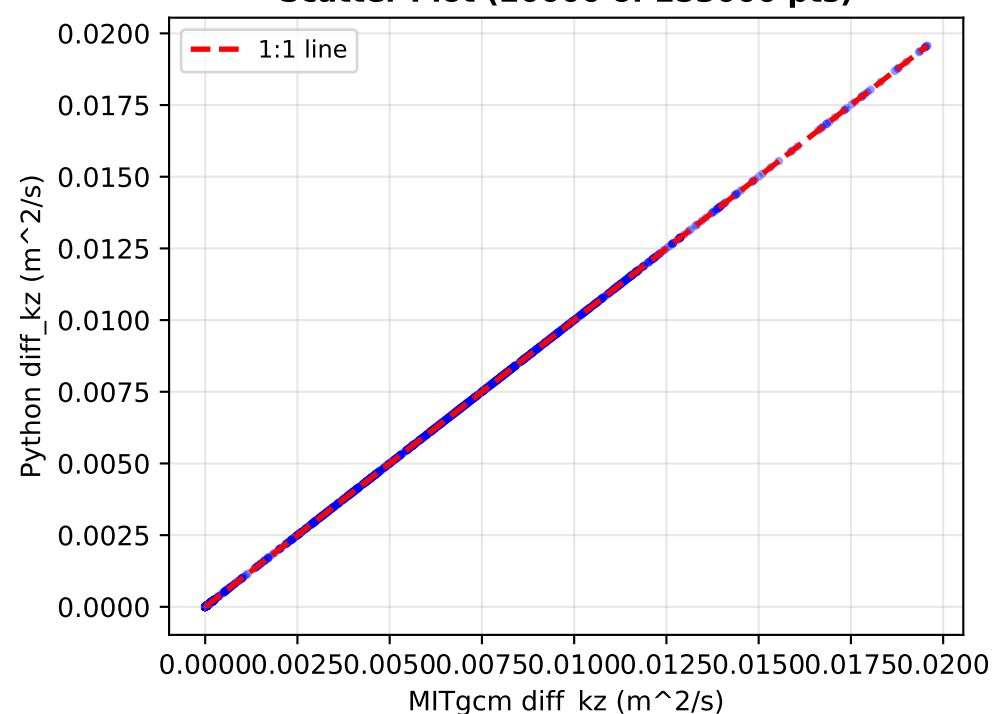


diff_kz (Vertical eddy diffusivity (K_h)) -- MITgcm vs Python

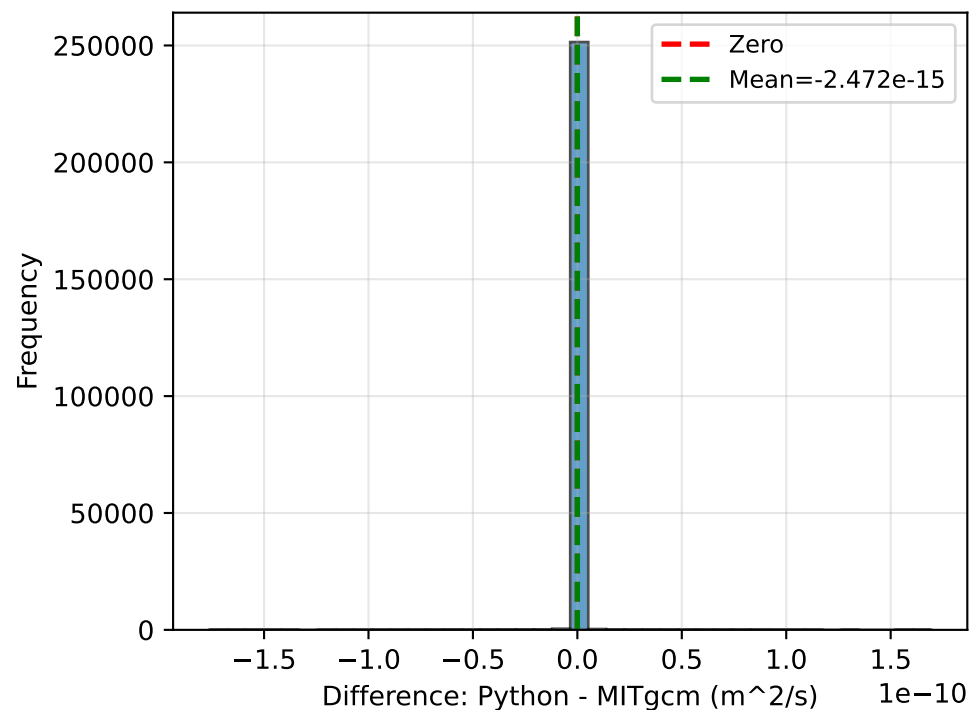
MITgcm Distribution



Scatter Plot (20000 of 253000 pts)



Difference Distribution



diff_kz Statistics:

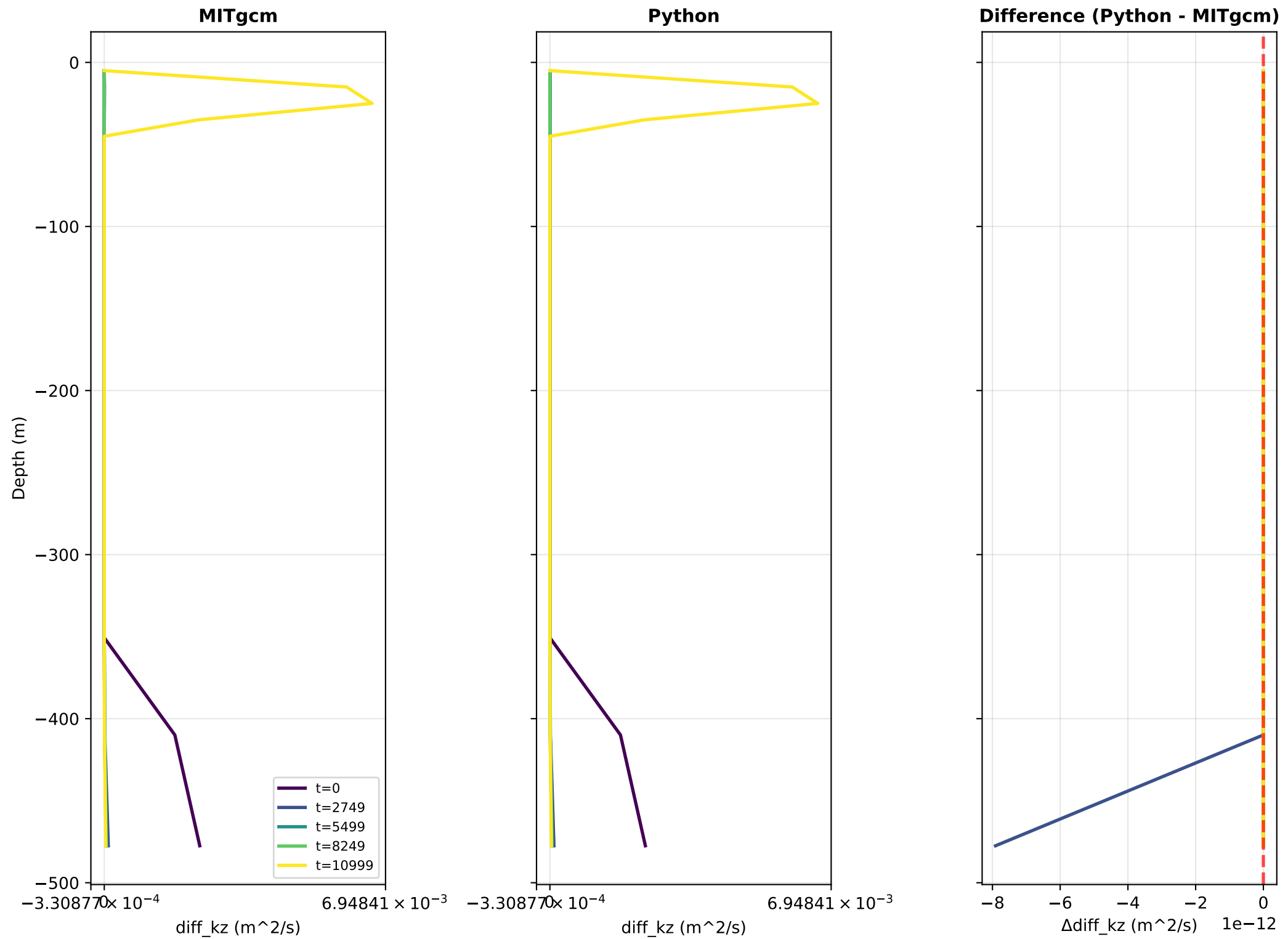
Median |diff|: 0.0000e+00
P95 |diff|: 3.1995e-17
Max |diff|: 1.7627e-10

Median rel: 0.0000e+00
P95 rel: 1.6326e-11
Max rel: 7.9362e-07

Fraction >1% rel: 0.0000%
Points compared: 253000

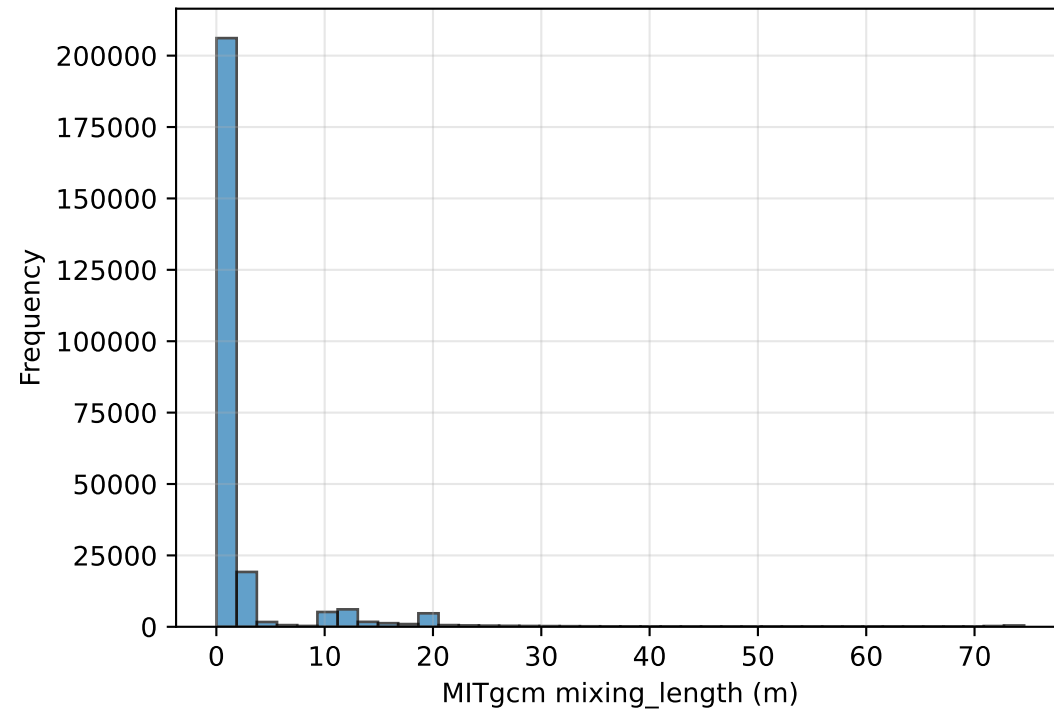
Assessment:
EXCELLENT

diff_kz Vertical Profiles (Vertical eddy diffusivity (K_h))

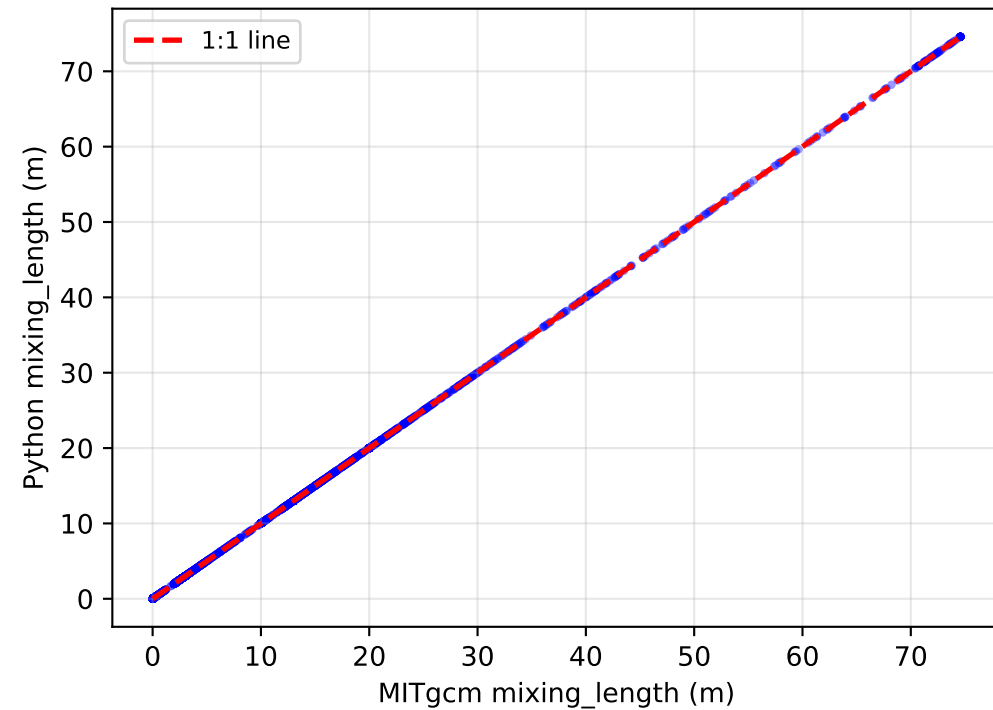


mixing_length (GGL90 mixing length (L)) -- MITgcm vs Python

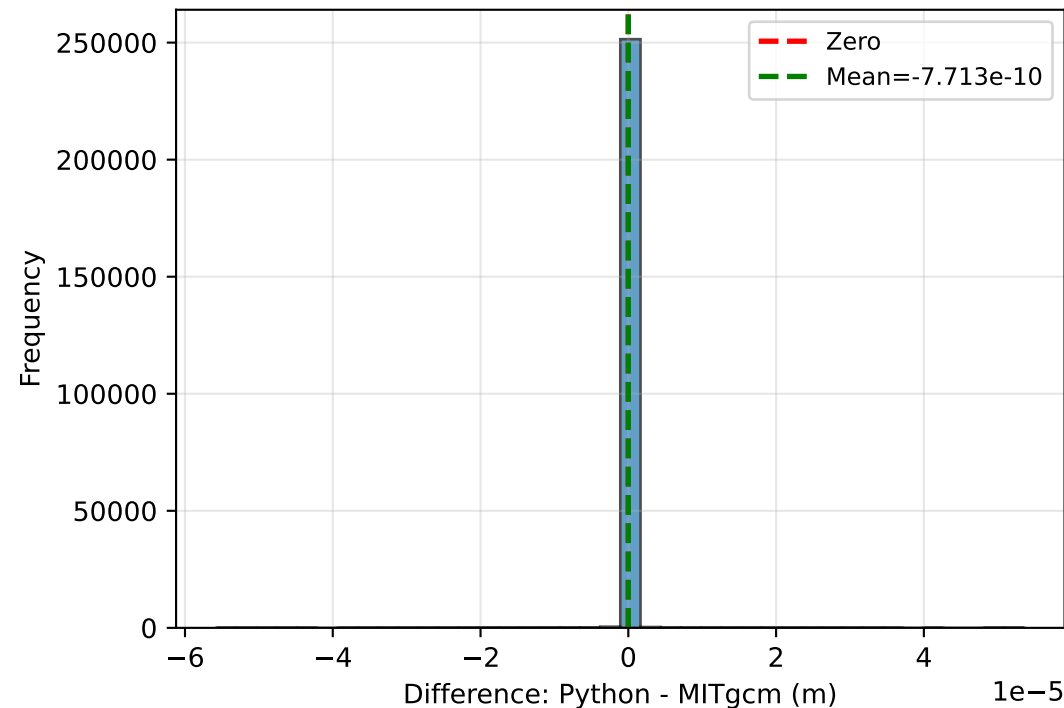
MITgcm Distribution



Scatter Plot (20000 of 253000 pts)



Difference Distribution



mixing_length Statistics:

Median |diff|: 2.5778×10^{-15}
P95 |diff|: 4.5353×10^{-12}
Max |diff|: 5.5741×10^{-5}

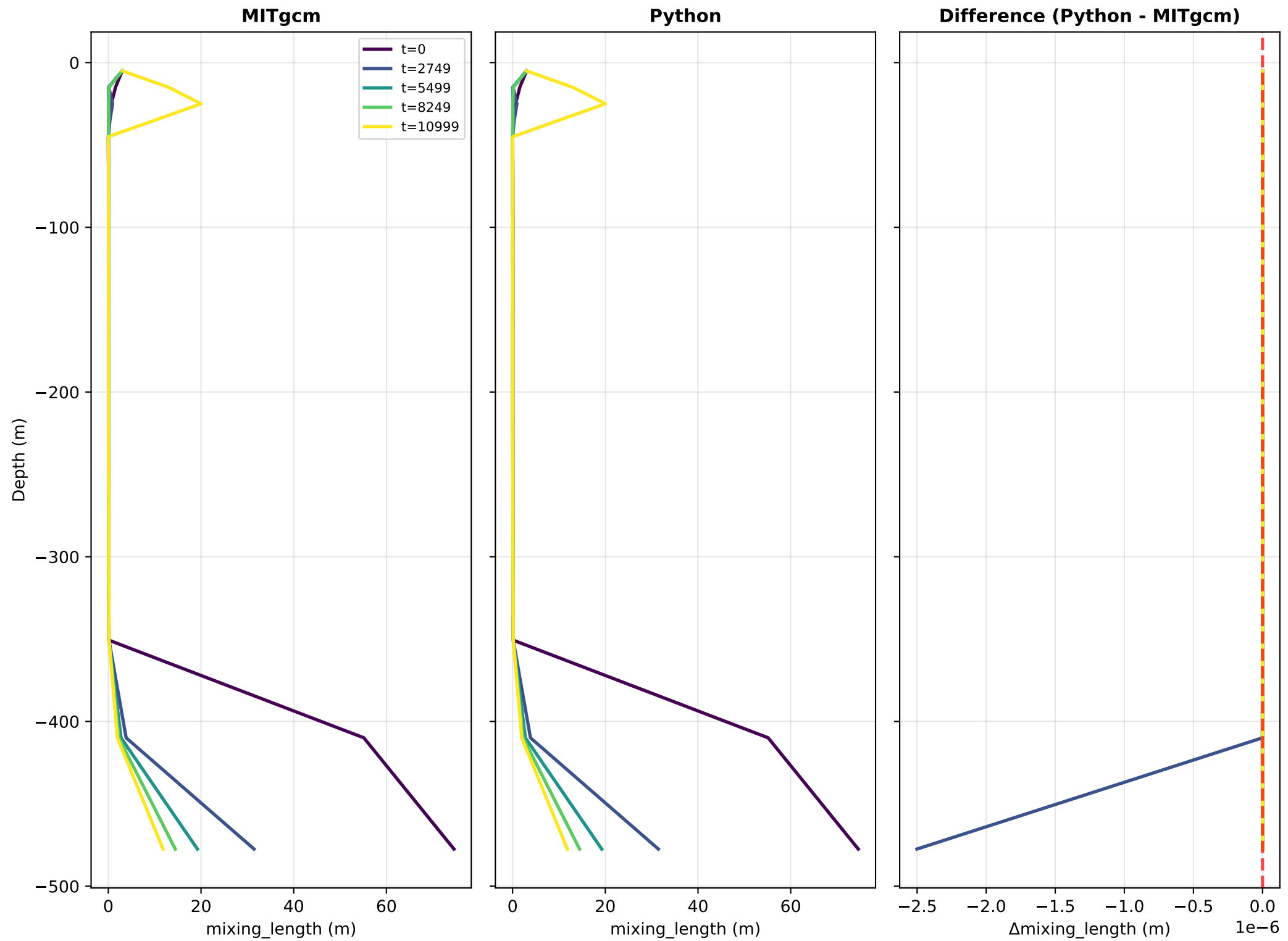
Median rel: 3.8561×10^{-15}
P95 rel: 1.6326×10^{-11}
Max rel: 7.9362×10^{-7}

Fraction >1% rel: 0.0000%
Points compared: 253000

Assessment:

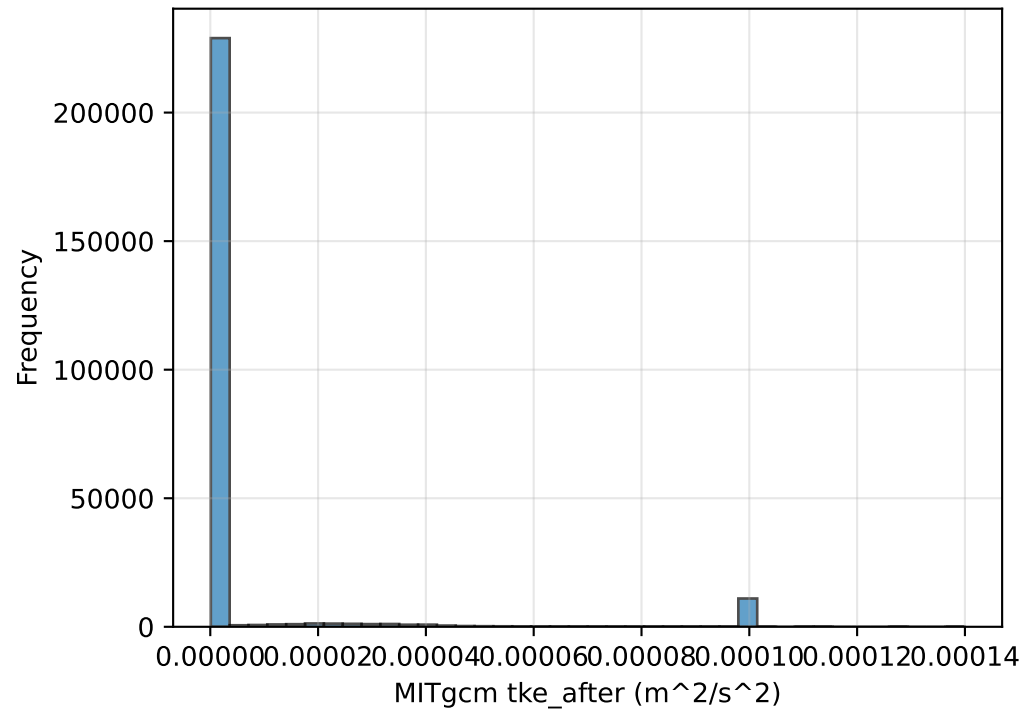
EXCELLENT

mixing_length Vertical Profiles (GGL90 mixing length (L))

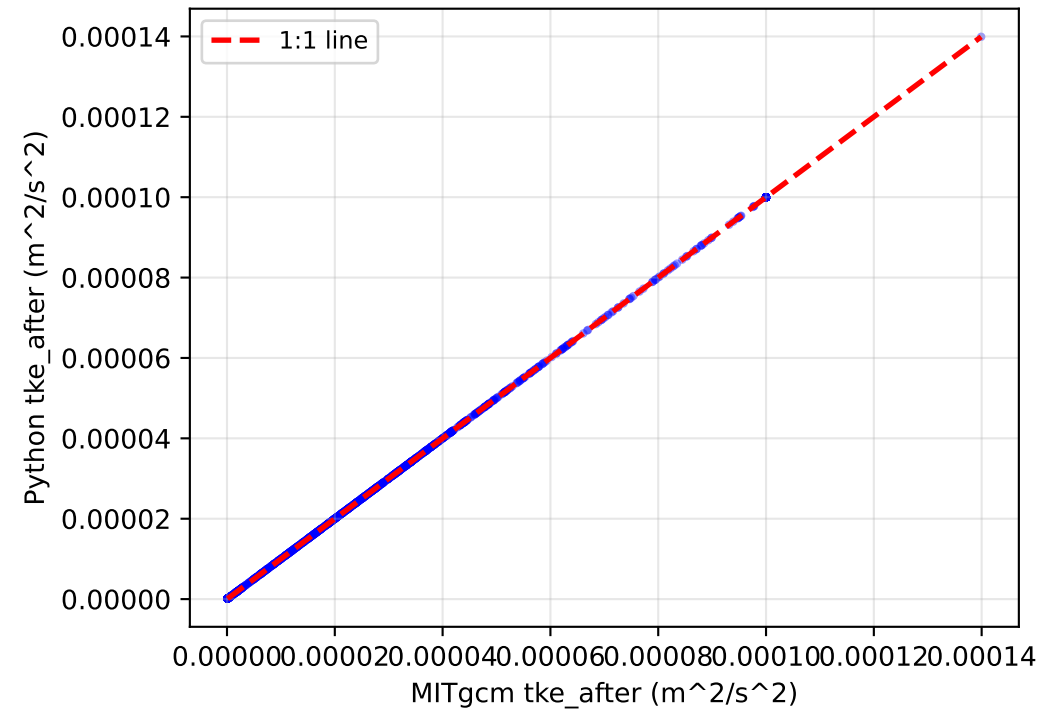


tke_after (TKE at t+1 (single-step replay)) -- MITgcm vs Python

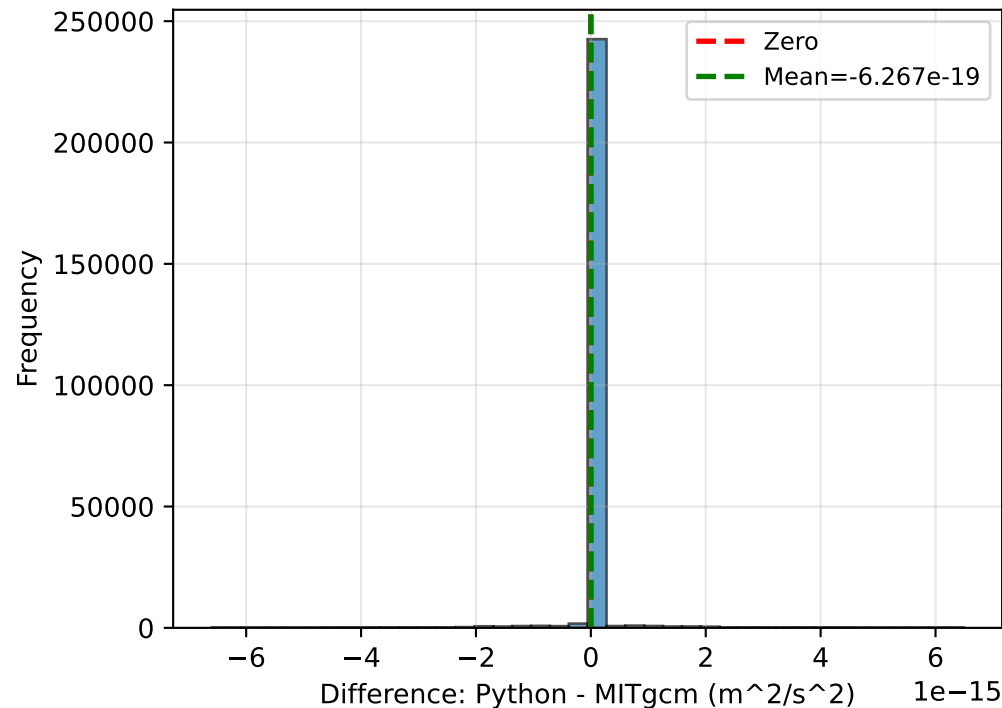
MITgcm Distribution



Scatter Plot (20000 of 253000 pts)



Difference Distribution



tke_after Statistics:

Median |diff|: 0.0000e+00
P95 |diff|: 1.6139e-17
Max |diff|: 6.6155e-15

Median rel: 0.0000e+00
P95 rel: 1.8797e-12
Max rel: 5.1386e-09

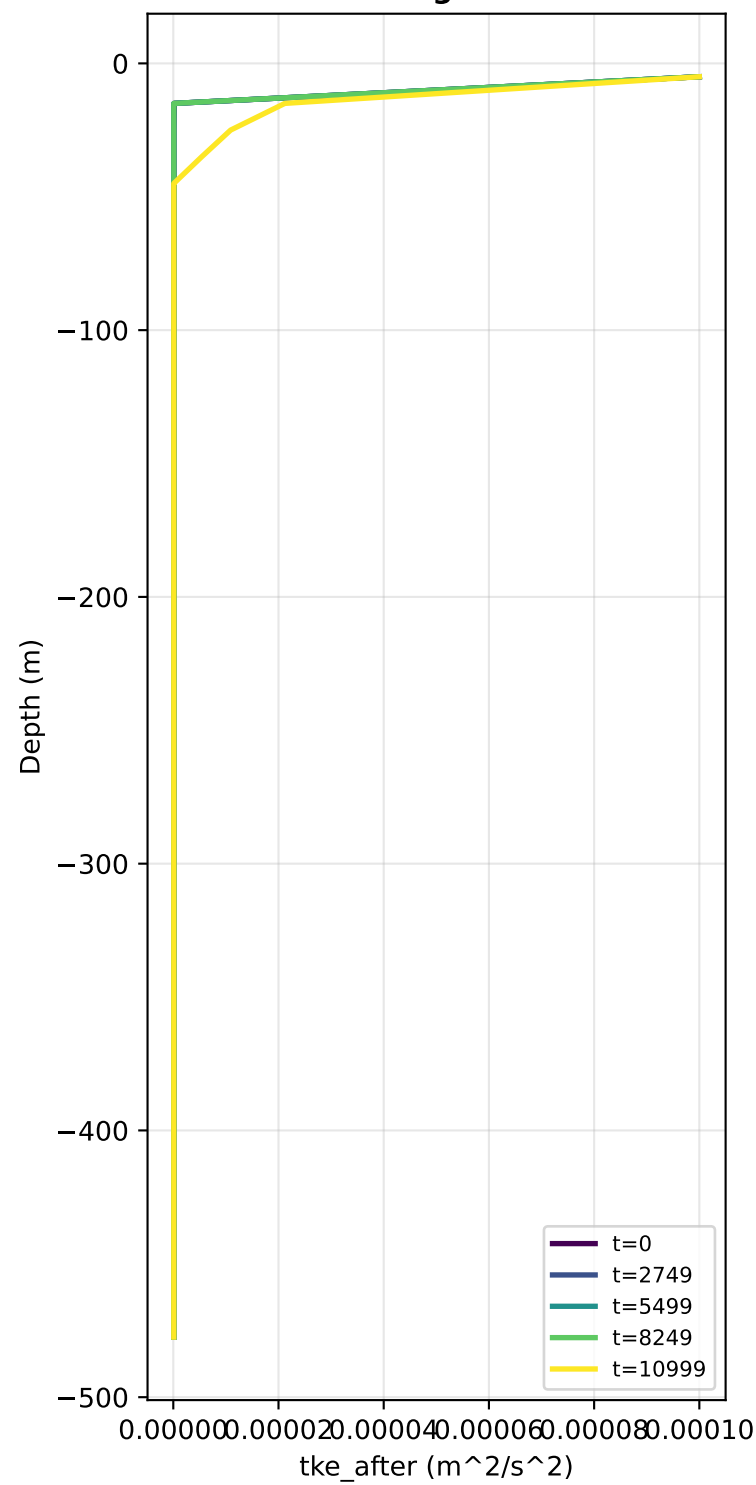
Fraction >1% rel: 0.0000%
Points compared: 253000

Assessment:

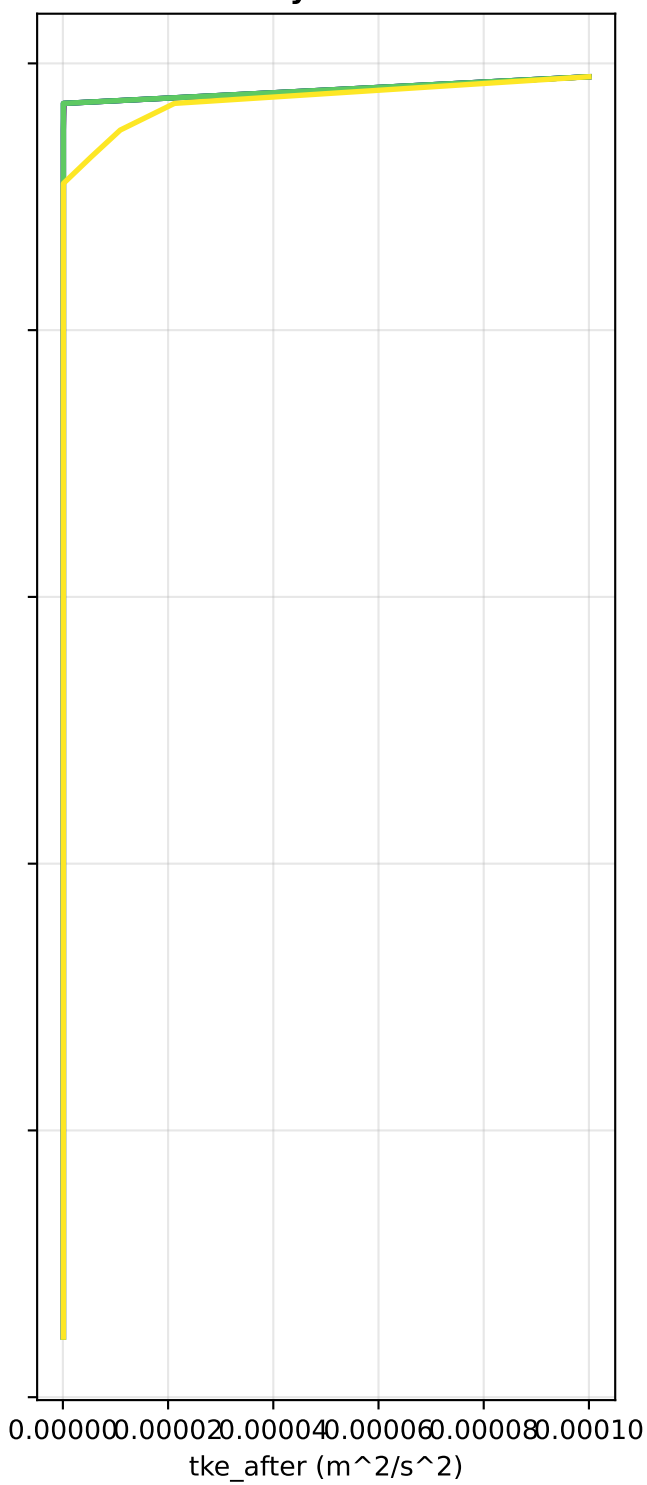
EXCELLENT

tke_after Vertical Profiles (TKE at t+1 (single-step replay))

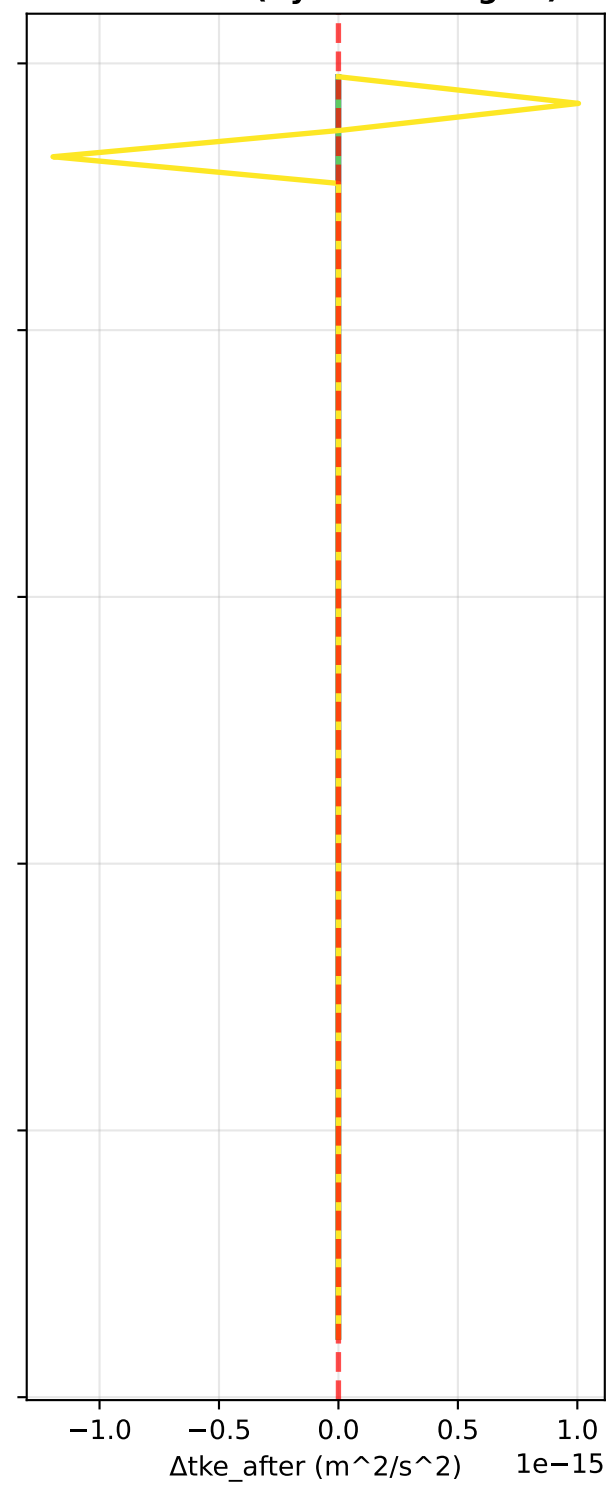
MITgcm



Python



Difference (Python - MITgcm)



Validation Summary & Assessment

OVERALL VALIDATION RESULT:

Status: ☒ PASS

COMPONENT RESULTS:

visc_az: ☒ EXCELLENT
Median rel error: 0.0000% Max |diff|: 1.7627e-09
Fraction >1% rel: 0.0000%

diff_kz: ☒ EXCELLENT
Median rel error: 0.0000% Max |diff|: 1.7627e-10
Fraction >1% rel: 0.0000%

mixing_length: ☒ EXCELLENT
Median rel error: 0.0000% Max |diff|: 5.5741e-05
Fraction >1% rel: 0.0000%

tke_after: ☒ EXCELLENT
Median rel error: 0.0000% Max |diff|: 6.6155e-15
Fraction >1% rel: 0.0000%

PROVENANCE:

UUID Match: ☒ Yes
MITgcm UUID: 1c8c7e52-3041-4f49-ae2b-841705d2d7a3
Python UUID: 1c8c7e52-3041-4f49-ae2b-841705d2d7a3

ACCEPTANCE CRITERIA (mirrors generate_kpp_validation_report.py's own):

☐ EXCELLENT: median rel error < 0.1%
☐ GOOD: median rel error < 1.0%
☒ CHECK: median rel error >= 1.0% -- may be a real, already-characterized gap (see closed_issues.md), not necessarily a new defect; this report states the number, not the cause.